Swiggy Sales Data Analysis

SQL Queries for Business Insights

Customer Analysis | Restaurant Performance | Delivery Metrics



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This project showcases a comprehensive MySQL-based data analysis of Swiggy, one of India's leading online food delivery platforms. Leveraging a structured relational database, the analysis explores key datasets involving customers, restaurants, orders, delivery partners, and ratings.



The objective is to demonstrate how data-driven decision-making can be
supported by relational database systems in a real-world business context. This project is ideal for stakeholders looking to understand how SQL can be used for business intelligence (BI), performance analytics, and operational optimization in the food tech industry.

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AGENDA

Customer Behaviour Analysis

Restaurant Performance Metrics

Delivery Partner & Order Trends

Location Based Analysis

Order Timeline



1. Display all customers who live in 'Delhi'.

```sql
select name, city
 from customers
 where city = 'Delhi';
```

	name	city
▶	Rohini Verma	Delhi
	Manish Kumar	Delhi
	Sonali Mishra	Delhi



2. Find the average rating of all restaurants in 'Mumbai'.

```sql
select city,  round(avg(coalesce(rating)),2)
from restaurants
where city = 'Mumbai'
group by 1;
```

	city	average_rating
▶	Mumbai	4.30



3. List all customers who have placed at least one order.

```sql
SELECT DISTINCT
    c.*
FROM
    customers c
        JOIN
    orders o ON c.customer_id = o.customer_id;
```

customer_id	name	email	phone_number	city	address
1	Amit Sharma	amit.sharma@gmail.com	9876543210	Mumbai	A-12, Andheri West
2	Rohini Verma	rohini.verma@yahoo.com	9823456789	Delhi	B-23, Saket
3	Rajesh Gupta	rajesh.gupta@gmail.com	NULL	Mumbai	C-34, Colaba
4	Sneha Mehta	sneha.mehta@gmail.com	9812345678	Pune	NULL
5	Manish Kumar	NULL	9834567890	Delhi	D-45, Lajpat Nagar
6	Priya Singh	priya.singh@hotmail.com	9845678901	Kolkata	E-56, Salt Lake
7	Vikas Reddy	vikas.reddy@gmail.com	9878901234	Chennai	F-67, T. Nagar
8	Anjali Patel	anjali.patel@yahoo.com	9890123456	Ahmedabad	NULL
9	Suresh Nair	NULL	9801234567	Bangalore	G-78, Koramangala
10	Kavita Deshmukh	kavita.deshmukh@gmail.com	NULL	Pune	H-89, Hinjewadi
11	Vivek Bhatt	vivek.bhatt@gmail.com	9810987654	Indore	I-90, Vijay Nagar
12	Meera Joshi	NULL	9823567890	Jaipur	J-11, Malviya Nagar



4. Display the total number of orders placed by each customer.

```sql
SELECT
    customer_id, COUNT(*) AS total_orders
FROM
    orders
GROUP BY customer_id;
```

customer_id	total_orders
1	2
2	3
3	3
4	2
5	4



5. Find the total revenue generated by each restaurant.

```
select r.name, r.restaurant_id, sum(o.total_amount) as revenue_generated
from restaurants r join orders o on
r.restaurant_id= o.restaurant_id
group by 2;
```

name	restaurant_id	revenue_generated
Biryani House	3	5300.00
Taste of Punjab	5	600.00
Spice of India	1	1100.00
Coastal Delight	7	2100.00
Tandoori Flames	2	1200.00
Curry Pot	4	3200.00
Veggie Delight	8	1600.00
Gujarat Express	9	2550.00
Royal Biryani	6	650.00
Punjabi Tadka	11	900.00
Flavours of Ben...	12	4050.00



6. Find the top 5 restaurants with the highest average rating.

```
select  name, avg(rating) from restaurants
group by 1
having avg(rating) > 4.35
order by 2 limit 5;
```

name	avg(rating)
Andhra Spice	4.400000
The Great Indian Thali	4.400000
Udupi Palace	4.400000
Spice of India	4.500000
Rajasthani Rasoi	4.500000



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7. Display all customers who have never placed an order.

```
select c.customer_id, c.name  from customers c
left join orders o on  c.customer_id = o.customer_id
where o.customer_id is null;
```

customer_id	name
24	Sonal Kaur
25	Vivek Malhotra
26	Divya Iyer
27	Rakesh Yadav
28	Mona Sharma
29	Sudha Pillai
30	Gaurav Khanna



8. Find the number of orders placed by each customer in 'Mumbai'.

```
select (c.customer_id), c.name, c.city, count(o.order_id) as order_count from customers c
join orders o on c.customer_id = o.customer_id
group by c.customer_id
having city = "mumbai";
```

customer_id	name	city	order_count
1	Amit Sharma	Mumbai	2
3	Rajesh Gupta	Mumbai	3
19	Arjun Desai	Mumbai	2
23	Ravi Singh	Mumbai	2



9. Display all orders placed in the last 30 days.

```sql
select * from orders
where order_date >= '2024-08-15' - interval 10 day;
```

order_id	customer_id	restaurant_id	order_date	total_amount	status
1	1	3	2024-08-01 00:00:00	750.00	Completed
2	2	5	2024-08-02 00:00:00	600.00	Completed
3	3	1	2024-08-04 00:00:00	0.00	Cancelled
4	4	7	2024-08-01 00:00:00	850.00	Completed
5	5	2	2024-08-03 00:00:00	1200.00	Completed
7	6	8	2024-08-03 00:00:00	950.00	Completed
9	8	6	2024-08-02 00:00:00	650.00	Completed
11	10	12	2024-08-01 00:00:00	900.00	Completed
12	2	13	2024-08-04 00:00:00	550.00	Completed
13	3	14	2024-08-05 00:00:00	750.00	Completed



OBSERVATIONS AND SUGGESTIONS

OBSERVATIONS	SUGGESTIONS
• Orders have been placed in the last 10 days.	• Monitor peak traffic hours using time-based order analysis • Optimize delivery logistics by allocating more riders during peak hours • Launch flash deals during high-traffic periods to increase conversion • Implement personalized push notifications to re-engage recent customers • Analyze order patterns to forecast inventory needs more accurately • Introduce time-bound discounts to create urgency and drive more orders

10. List all delivery partners who have completed more than 1 delivery.

```sql
select dp.partner_id, dp.name, dp.city, count(od.order_delivery_id) as order_more_than_one
from deliverypartners dp
left join orderdelivery od
on dp.partner_id = od.partner_id
group by 1,2
having count(od.order_delivery_id)>1;
```

partner_id	name	city	order_more_than_one
1	Amit Sharma	Mumbai	2
2	Ravi Kumar	Delhi	5
3	Priya Patel	Bengaluru	3
4	Suresh Reddy	Hyderabad	6
5	Anita Desai	Chennai	4

Result 5



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11. Find the customers who have placed orders on exactly three different days.

```sql
SELECT c.customer_id, c.name
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
GROUP BY 1,2
HAVING COUNT(DISTINCT DATE(order_date)) = 3;
```

customer_id	name
2	Rohini Verma
6	Priya Singh
8	Anjali Patel
14	Nidhi Saxena
15	Ashok Kumar



12. Find the delivery partner who has worked with the most different customers.

```sql
select dp.partner_id, dp.name, dp.city, count(o.order_id) as different_customers from deliverypartners dp
join  orderdelivery o on dp.partner_id = o.partner_id
group by 1
order by different_customers desc
limit 1;
```

partner_id	name	city	different_customers
4	Suresh Reddy	Hyderabad	6



OBSERVATIONS AND SUGGESTIONS



OBSERVATIONS	SUGGESTIONS
Some delivery partners have multiple successful deliveries.	• Implement a tiered incentive program for top-performing partners. Use their delivery data to train or mentor new hires.
A few partners serve more unique customers.	• Analyze their delivery areas to identify underserved zones. Use them for pilot testing hyperlocal delivery campaigns.

13. Identify customers who have the same city and have placed orders at the same restaurants, but on different dates.

```sql
select distinct c1.name as first_customer_name, c2.name as repeated_customer_name, c1.city, o1.restaurant_id, o1.customer_id as first_customer_id,
o2.customer_id as repeated_customer_id, o1.order_date as first_orderdate,
o2.order_date as repeat_orderdate
from orders o1
join orders o2
on o1.restaurant_id= o2.restaurant_id
and o1.order_date<> o2.order_date
and o1.customer_id < o2.customer_id
join customers c1
on o1.customer_id = c1.customer_id
join customers c2
on o1.customer_id = c2.customer_id
and c1.city = c2.city;
```



first_customer_name	repeated_customer_name	city	restaurant_id	first_customer_id	repeated_customer_id	first_orderdate	repeat_orderdate
Amit Sharma	Amit Sharma	Mumbai	3	1	5	2024-08-01 00:00:00	2024-08-04 00:00:00
Amit Sharma	Amit Sharma	Mumbai	3	1	18	2024-08-01 00:00:00	2024-08-05 00:00:00
Amit Sharma	Amit Sharma	Mumbai	3	1	5	2024-08-01 00:00:00	2024-08-07 00:00:00
Amit Sharma	Amit Sharma	Mumbai	4	1	20	2024-08-06 00:00:00	2024-08-03 00:00:00
Amit Sharma	Amit Sharma	Mumbai	4	1	11	2024-08-06 00:00:00	2024-08-11 00:00:00

OBSERVATIONS AND SUGGESTIONS



OBSERVATIONS	SUGGESTIONS
A majority of customers have placed at least one order.	🎯 Launch retention campaigns to boost order frequency among active users.
📊 Order frequency differs significantly across customers.	🔁 Introduce tiered loyalty programs to incentivize frequent ordering.
⛔ Some customers have never placed an order.	📩 Engage these users through personalized onboarding and targeted offers.
📅 Certain users place orders on exactly three unique days.	🧠 Analyze behavioral patterns within this group to test timing-based engagement strategies.
🌆 Customers from the same city order from the same restaurants on different dates.	👥 Encourage group ordering, referral programs, or social nudges like “Your friend ordered here.”

Thank You

By Amit Sharma

